

JOINT REVIRANT AND NRC SYMPOSIUM ON RESPIRATORY PATHOGENS

SARI SURVEILLANCE

28/04/2025

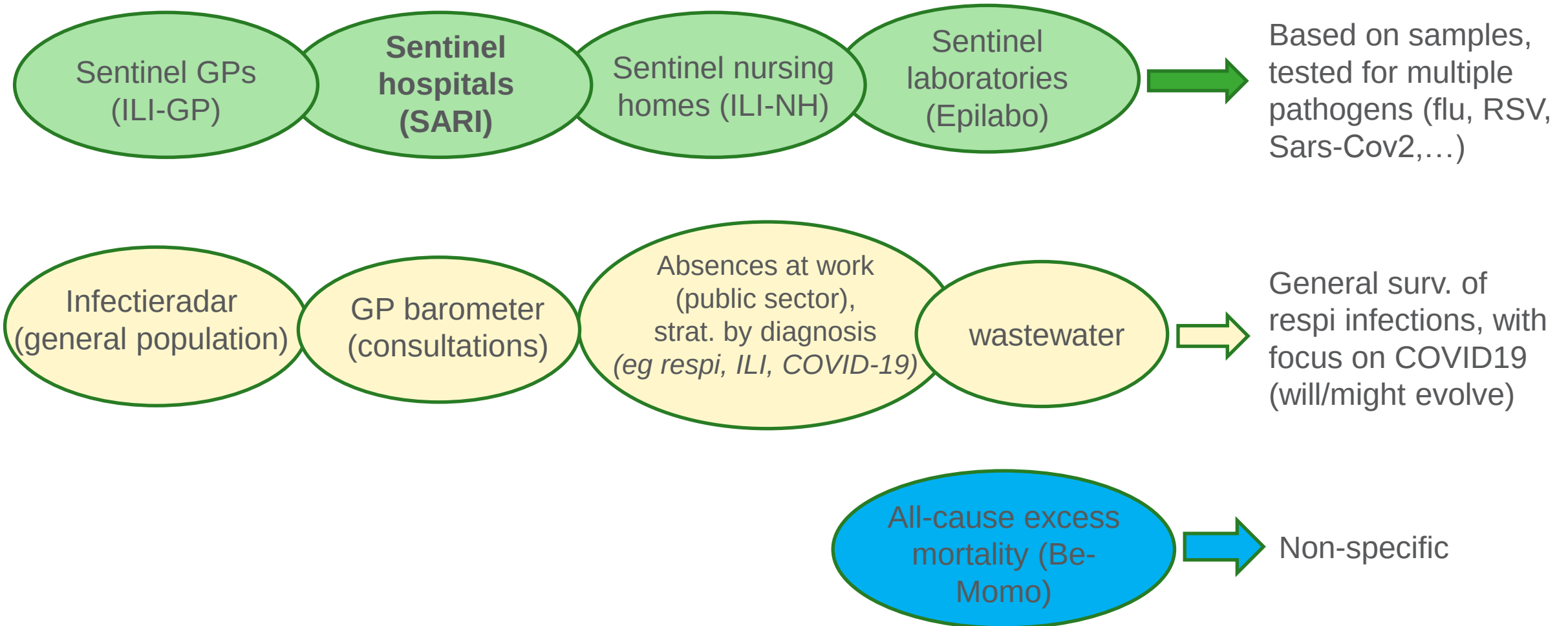
Outline

- Introduction
- Description
- Incidences
- Outputs
- Conclusion

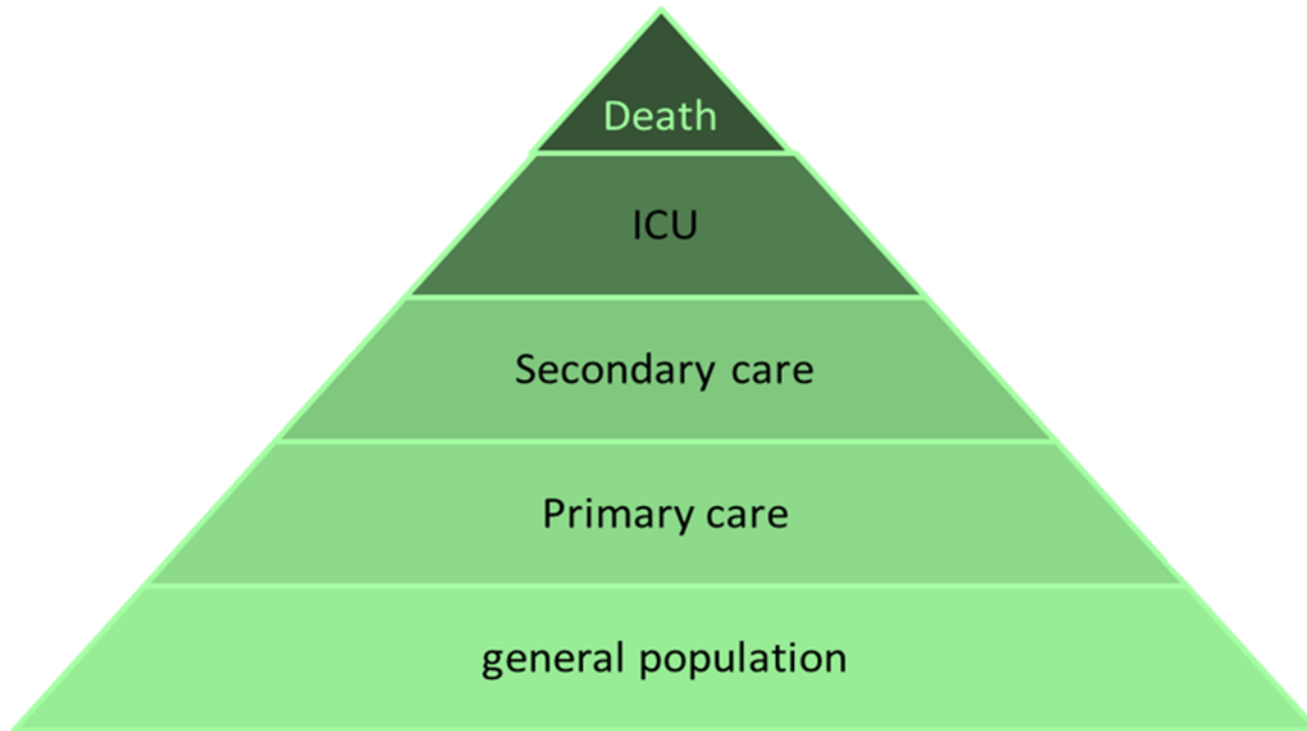
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Current systems for surveillance of respiratory infections



Respiratory infections surveillance pyramid



Objectives

- Describe stages of epidemic (intensity, duration, severity)
- Describe circulating pathogens
- Estimate impact (health services, mortality)
- Vaccine/immunization effectiveness analyses
- Selection of samples for sequencing and to test resistance

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SARI Surveillance – description

- **Severe Acute Respiratory Infection** Belgian SARI Surveillance system:

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System design	Sentinel (>< comprehensive)	Network of sentinel hospitals : Reference hospitals, high ICU capacity, geographical spread
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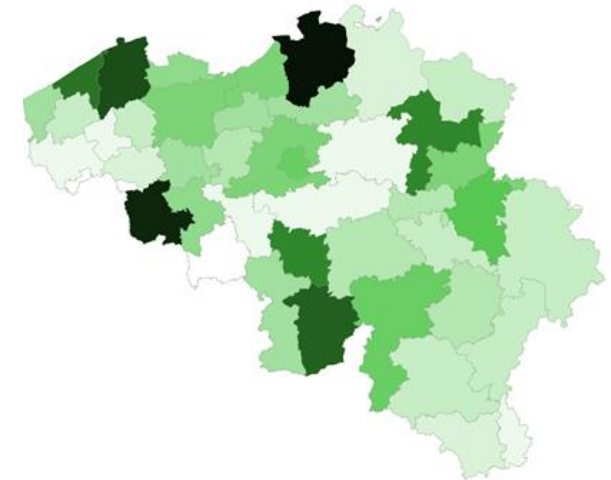
SARI Surveillance – description

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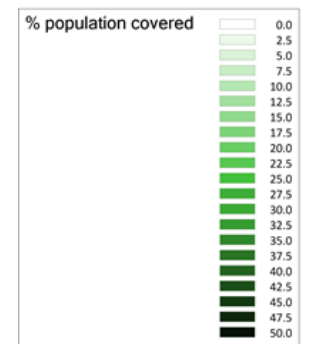
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- Geographical spread for representativeness

Flanders (5)	Wallonie (3)	Brussels (2)
Limburg Jessa Ziekenhuis	Namur CHU UCL namur (Mont-Godinne)(*pediatrics Dinant)	Brussels CHU St Pierre UZ Brussels
West-vlaanderen AZ Sint-jan Brugge	Hainaut Grand Hôpital de Charleroi Centre Hospitalier De Wallonie Picarde - Chwapi	
Antwerp Ziekenhuisnetwerk Antwerpen (Universitair Ziekenhuis Antwerpen)		
Oost-vlaanderen Universitair Ziekenhuis Gent		



*Catchment
population
~ 15% of the
belgian pop.*



SARI Surveillance – description

- Severe Acute Respiratory Infection Belgian SARI Surveillance system:

System design	Sentinel (>< comprehensive)	Network of sentinel hospitals : Reference hospitals, high ICU capacity, geographical spread
SARI case definition	Symptoms (>< diagnostic codes)	extended in 2023 to better identify respiratory infections other than influenza (eg SARS-Cov-2) and infections in different age groups (focus on RSV)

- Include all patients meeting clinical case definition
→ for each patient, send clinical data and nasopharyngeal swab
- exhaustive at the scale of the hospital

> 16 year old	<= 16 year old
*presenting with a respiratory illness with an acute onset (within the last 10 days) *requiring hospitalization (for 24h or more) for this specific reason. *presenting with ≥2 of the following signs of respiratory infection: • Fever ≥ 38°C (measured at the hospital) or history of fever (measured at home) • Cough • Signs of respiratory distress (including but not limited to dyspnoea) • Abnormal pulmonary auscultation	*presenting with a respiratory illness with an acute onset (within the last 10 days) *requiring hospitalization (for 24h or more) for this specific reason. *presenting with ≥2 of the following signs of respiratory infection: • Fever ≥ 38°C (measured at the hospital) or history of fever (measured at home) • Cough • Signs of respiratory distress (including but not limited to grunting or shortness of breath or polypnea/hypopnea or chest indrawing/retractions or nasal flaring) • Abnormal pulmonary auscultation • Apnea or cyanosis

SARI Surveillance – description

- **Severe Acute Respiratory Infection** Belgian SARI Surveillance system:

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Collection type	Active (>< passive)	Support with study coordinator - Data quality - Timeliness

SARI Surveillance – description

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SARI Surveillance – description

- Severe **A**cute **R**espiratory **I**nfection Belgian SARI Surveillance system:

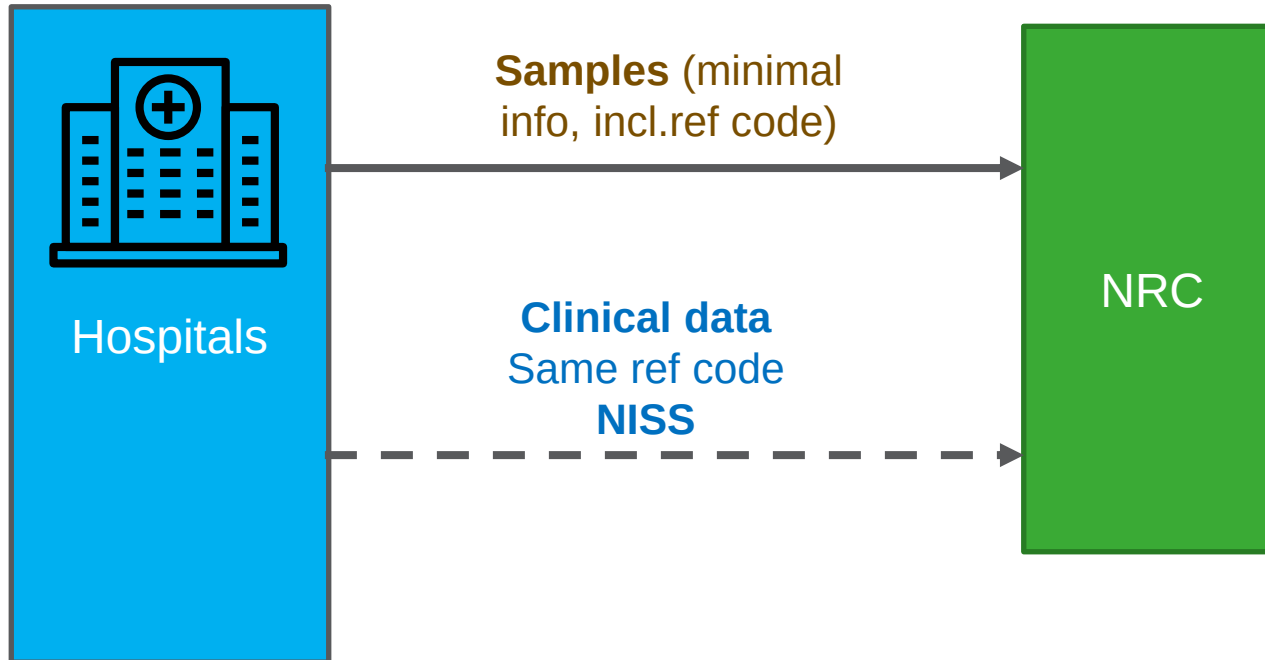
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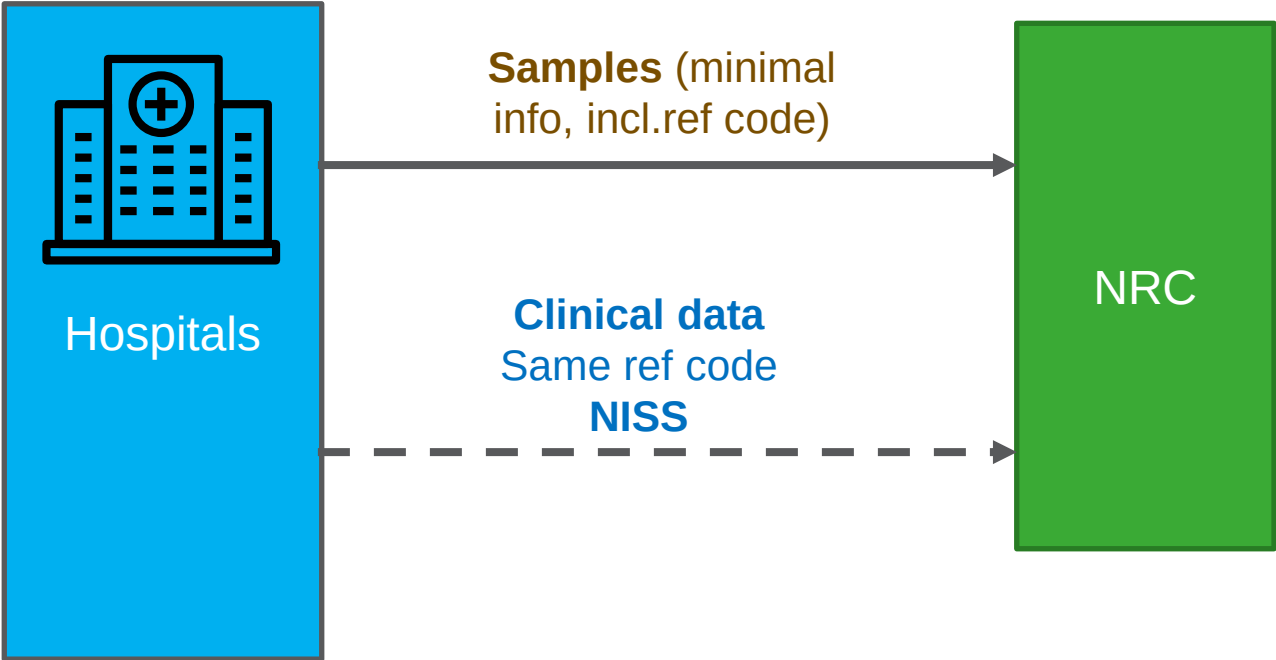
Standardized case identification in a timely manner

- **comparability** and pooled estimations
- **Weekly interpretation** of the epidemiological situation

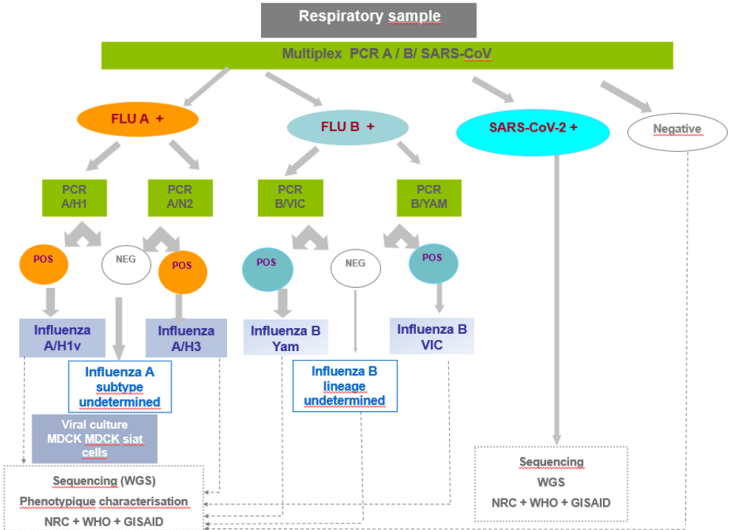
SARI Surveillance – description



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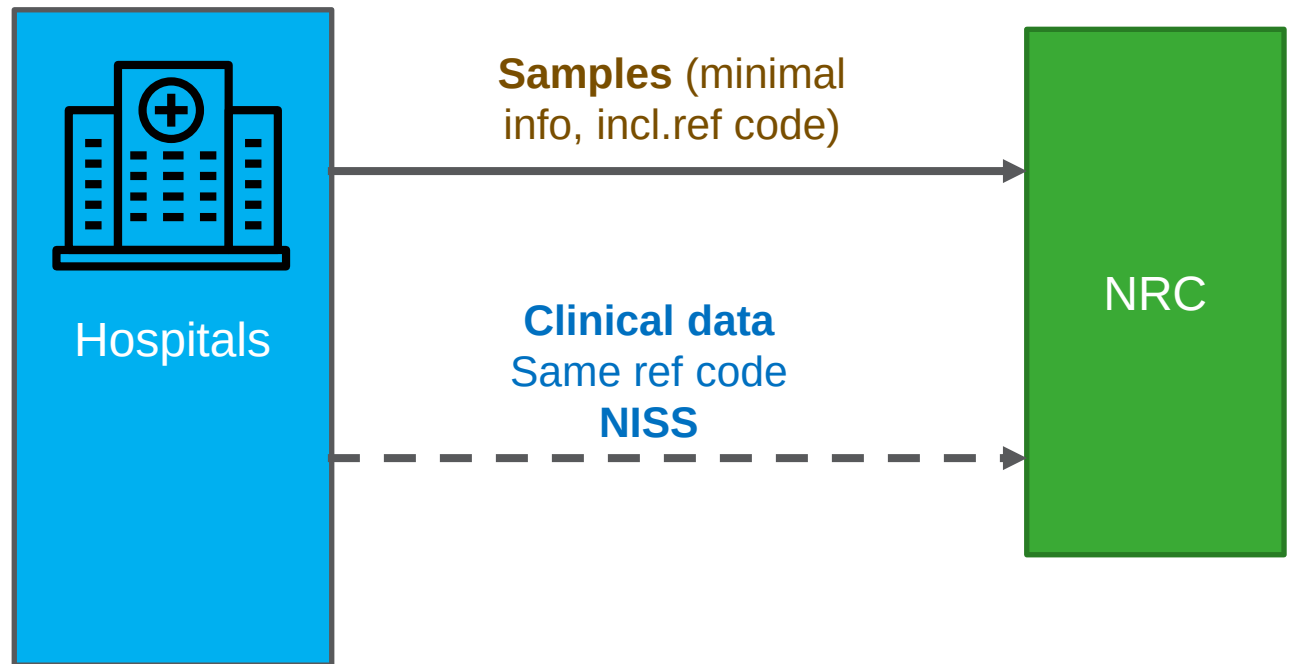


	SARI PROTOCOL	Edition : 2023-2034
		Annexes: 8
		P. : 1/57

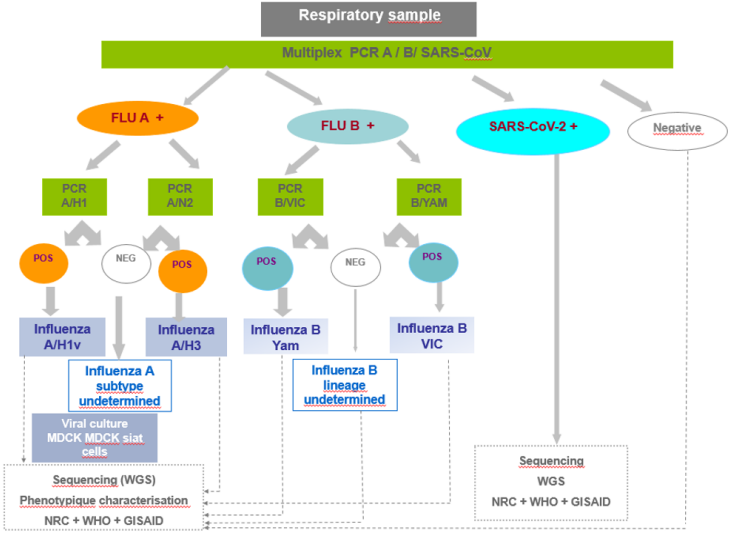


MIX 1		MIX 2		MIX 3		MIX 4	
RSV A	HEX	PIV 1	ROX	Co 229E	ROX	Boca	Cy5
RSV B	ROX	PIV 2	HEX	Co OC43	HEX	PIV 4	HEX
hMPV	Cy5	PIV 3	FAM	Co NL63	Cy5	Paraecho	ROX
EV	FAM	Adéno	Cy5	Rhino	FAM	EV-D68	FAM

SARI Surveillance – description



	SARI PROTOCOL	Edition : 2023-2034 Annexes: 8 P. : 1/57
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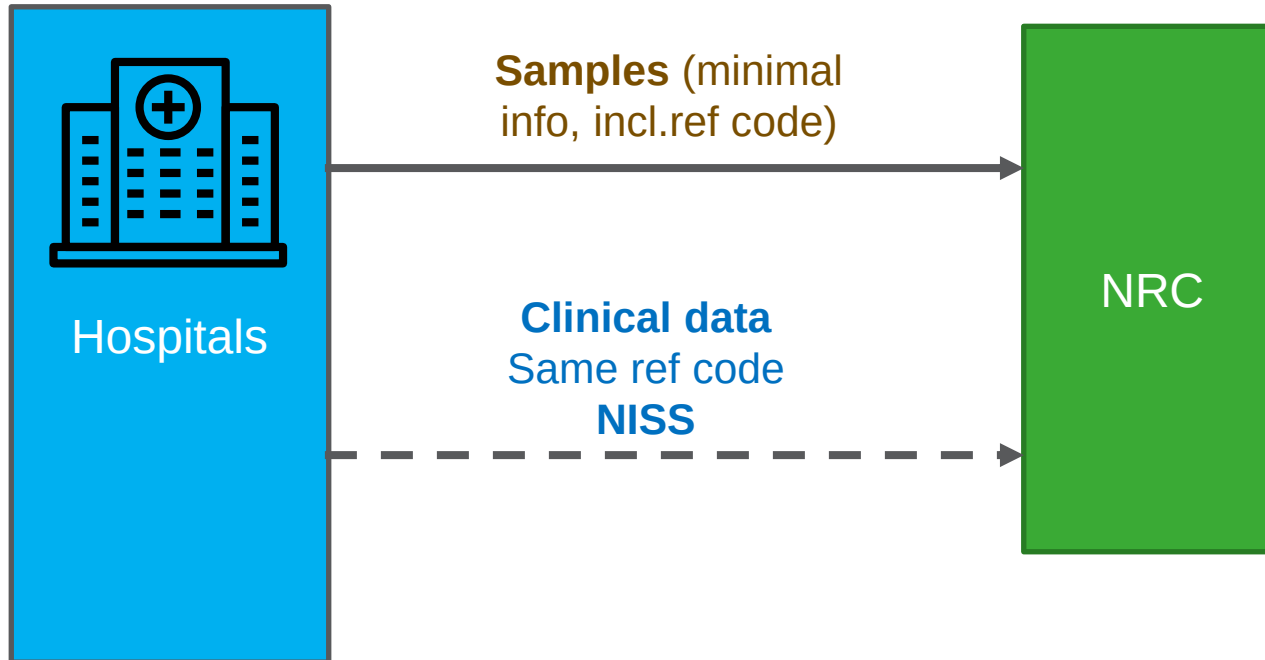


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hMPV	Cy5	PIV 3	FAM	Co NL63	Cy5	Paraecho	ROX
EV	FAM	Adéno	Cy5	Rhino	FAM	EV-D68	FAM

Extensive uniform sample testing in a retrospective manner

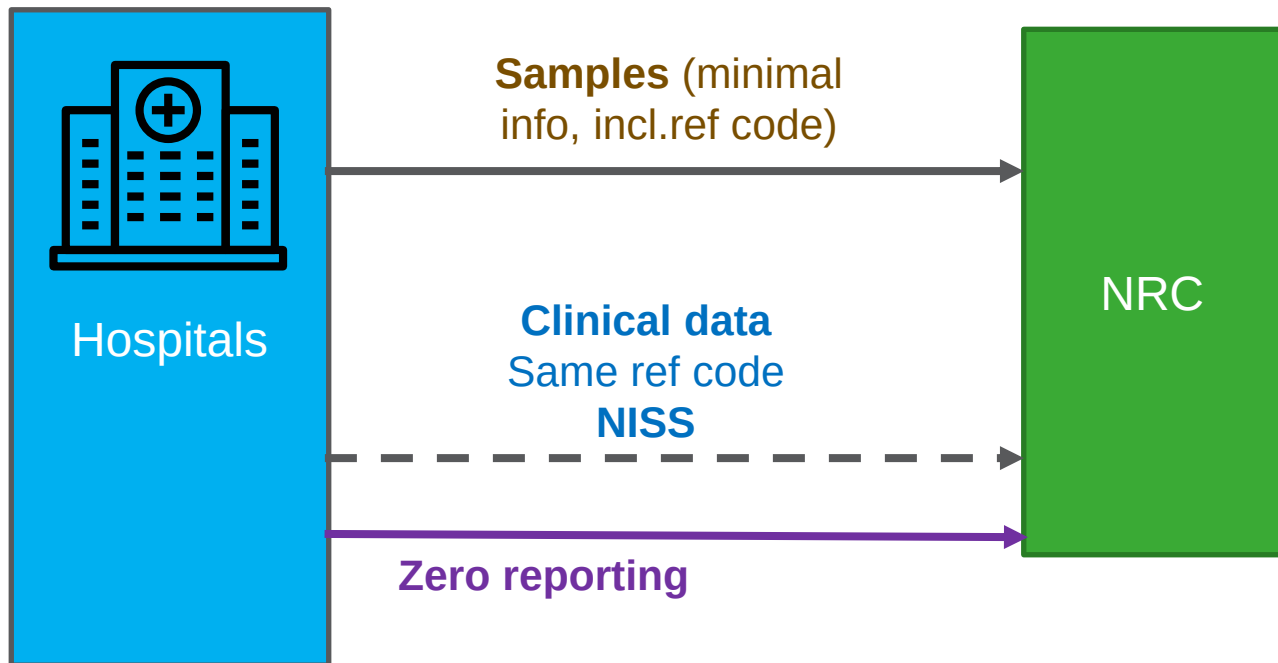
- comparability of test results
- All-round interpretation of the epidemiological situation

SARI Surveillance – description



- Collected separately :
 - Minimal information at admission
 - Detailed follow-up information at discharge
- Standardized questionnaire
- paper form → electronic questionnaire/batchupload
 - Built-in instructions
 - Structured default values
 - Integrated validation checks
 - Metadata template

SARI Surveillance – description



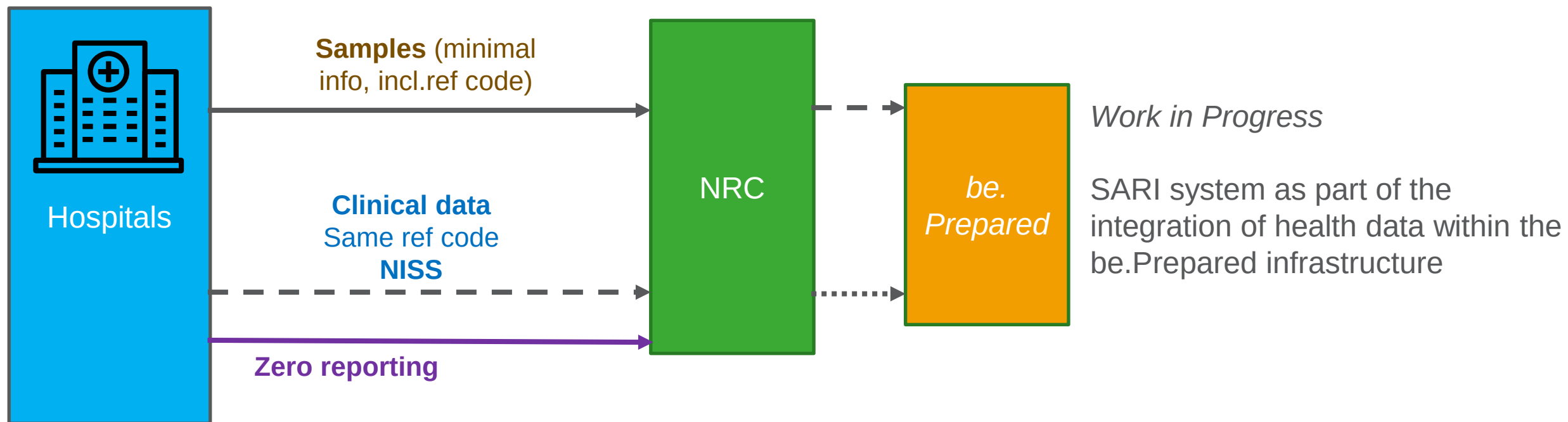
- Data processing
 - Cleaning steps
 - Validation steps
 - Merging
- Data analysis
 - Zero reporting
 - Systematic output



Systematic validated data collection/processing

- ➔ Increased data quality
- ➔ Detailed collection of the patient profiles

SARI Surveillance – description



Prospective health data integration

- ➔ Pandemic preparedness
- ➔ linkage with other datasets

Outline

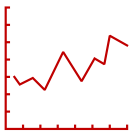
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SARI Surveillance – Incidences

- Necessary to “correct” absolute number of patients by size of the network
- Incidences are computed by dividing the numerator (number of cases) by a denominator (catchment population)
- Denominators are computed based on the market shares by hospitals
 - Market shares express the % of hospitalisations amongst hospitals in a given municipality by age groups

NIS-code	Post-code	Naam	Aantal verblijven	Klasse	Aantal verblijven	Erkenning	Benaming	Gemeente	Aantal verblijven	Markt-aandeel
11001	2630	Aartselaar	2.110	Jonger dan 15 jaar	146	009	ZIEKENHUISNETWERK ANTWERPEN (ZNA)	Antwerpen	21	14,4 %
11001	2630	Aartselaar	2.110	Jonger dan 15 jaar	146	026	AZ SINT-MAARTEN	Mechelen	7	4,8 %
11001	2630	Aartselaar	2.110	Jonger dan 15 jaar	146	099	GZA- ZIEKENHUIZEN	Antwerpen	41	28,1 %
11001	2630	Aartselaar	2.110	Jonger dan 15 jaar	146	300	UNIVERSITAIR ZIEKENHUIS ANTWERPEN	Edegem	50	34,2 %
11001	2630	Aartselaar	2.110	Jonger dan 15 jaar	146	322	UZ LEUVEN	Leuven	5	3,4 %
11001	2630	Aartselaar	2.110	Jonger dan 15 jaar	146	682	AZ MONICA	Antwerpen	17	11,6 %
11001	2630	Aartselaar	2.110	Tussen 15 en 64 jaar	849	009	ZIEKENHUISNETWERK ANTWERPEN (ZNA)	Antwerpen	111	13,1 %
11001	2630	Aartselaar	2.110	Tussen 15 en 64 jaar	849	026	AZ SINT-MAARTEN	Mechelen	20	2,4 %

- Both numerator and denominator are corrected for participation : N days and age groups
- Raw incidence = total number of cases/ catchment population



Pooled data assessment

- ➔ **Robust** and **consistent** interpretation
- ➔ **Comparability over time** (thresholds for epidemic assessment)

Outline

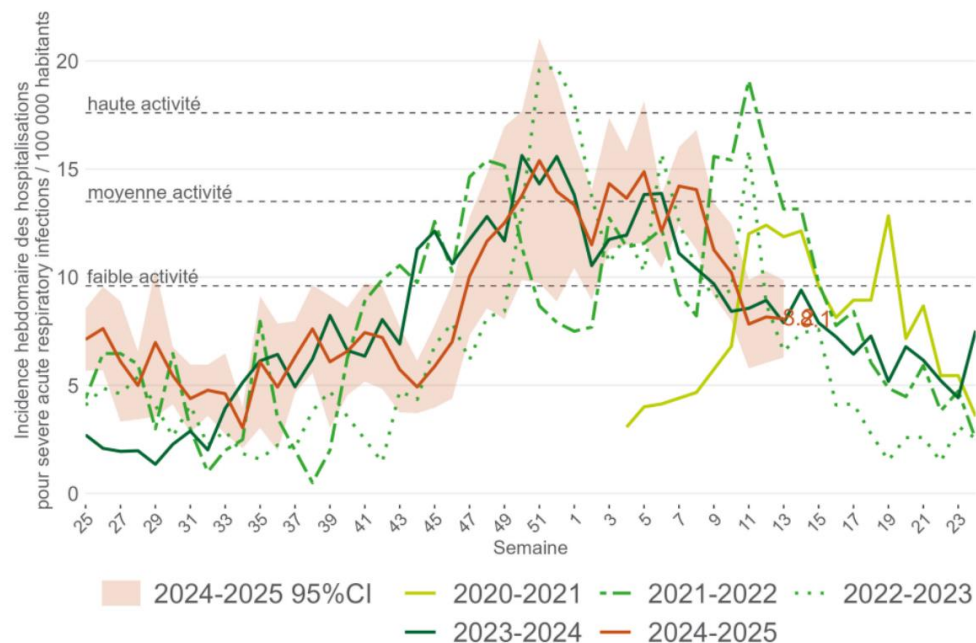
- Introduction
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 - Bulletin/Dashboard
 - Vaccine/Immunization effectiveness analysis
 - Respiradar
 - European Reporting
 - European Projects
- Conclusion

SARI Surveillance - outputs

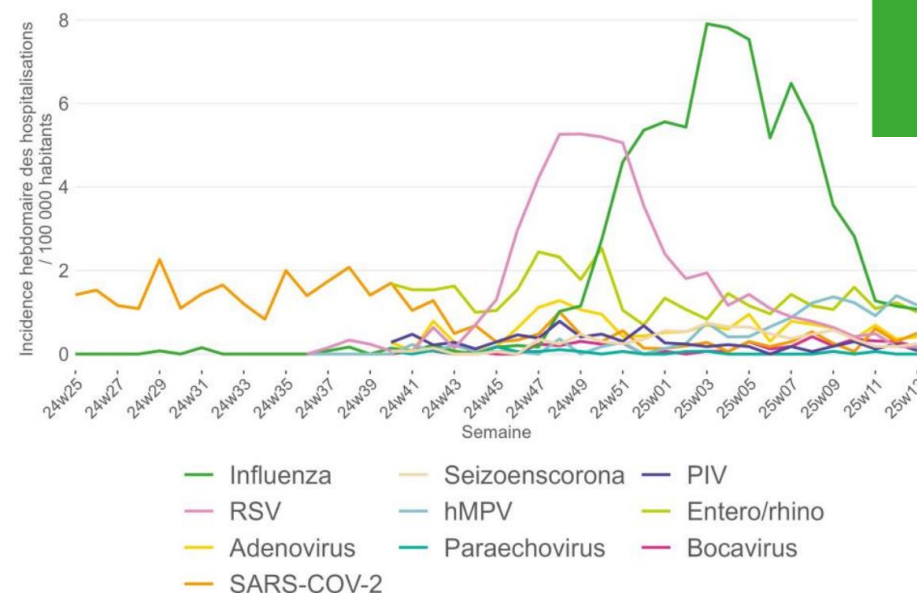
Bulletin infections respiratoires aiguës semaine 14-2025 - Publié 09/04/25 | sciensano.be



SARI incidence



SARI incidence per pathogen

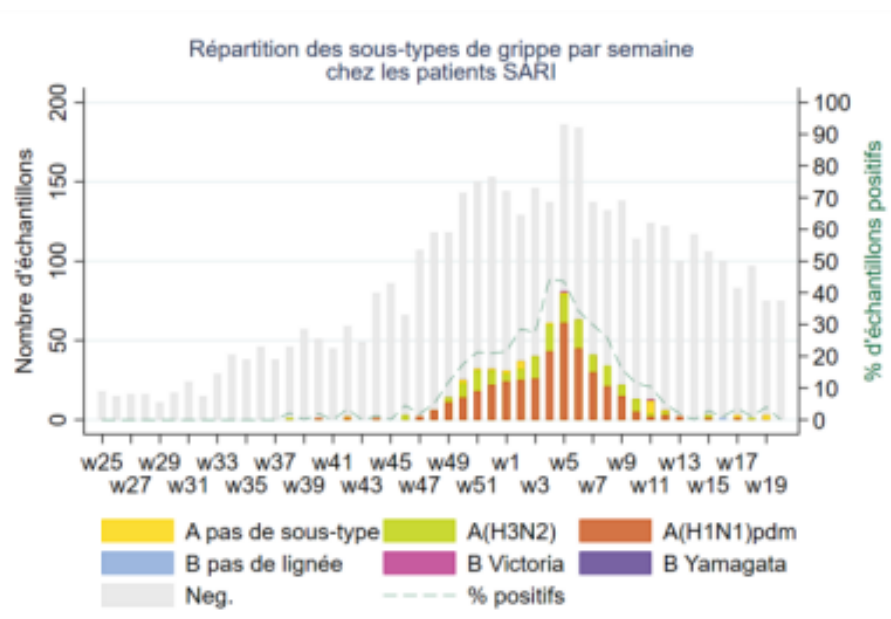


Bulletin hebdomadaire
infections respiratoires aiguës

Semaine 14 (31/03/2025 - 06/04/2025)

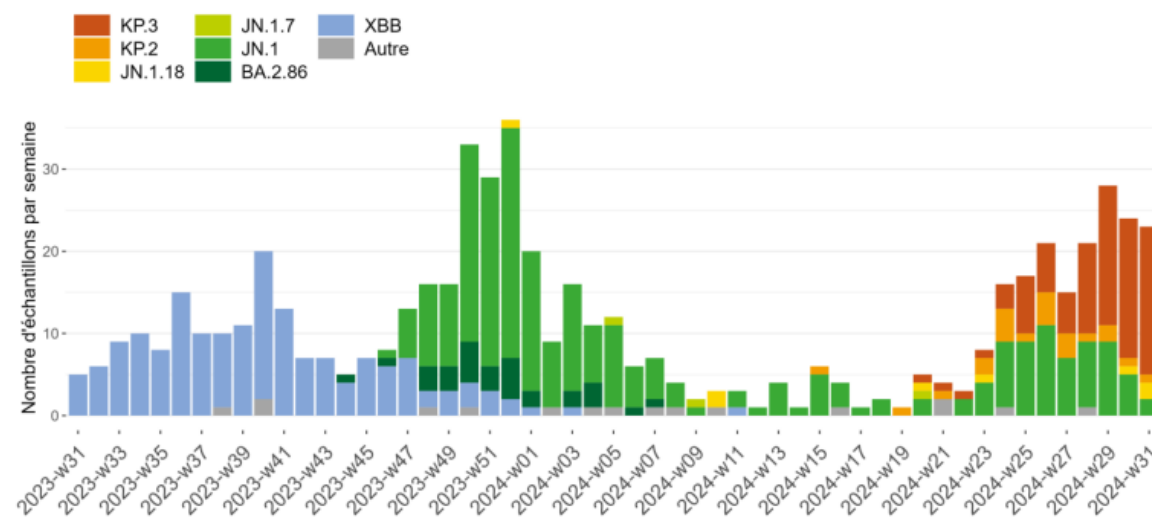
SARI Surveillance - outputs

Flu Subtypes



Sequencing data

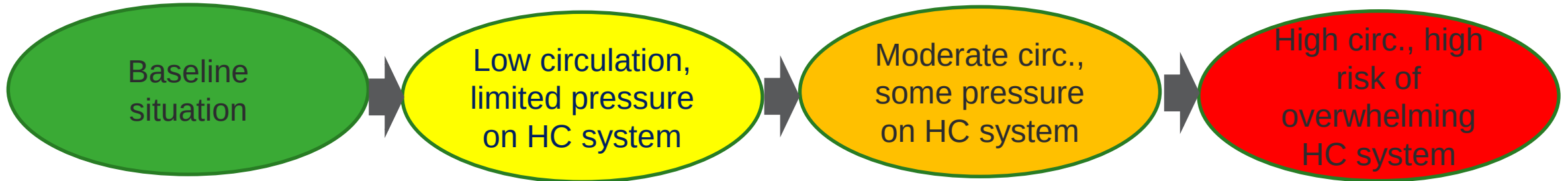
Nombre d'échantillons par variant identifiée dans le cadre de la surveillance du SARI en Belgique depuis le 31 juillet 2023, par date de prélèvement



- Dashboard <https://posit-cn.sciensano.be/epistat/dashboard/#respi>
- Vaccine/Immunization effectiveness analysis (*covid, flu, RSV**)
- Respiradar (RAG)

SARI Surveillance - outputs : Respi-radar

- Objectives: **Assess** the severity of the epidemiological situation of respiratory infections and **inform** public health preparedness and response
- Based on several indicators with their own thresholds
- **Risk levels** based on indicators reflecting the epidemiological situation –



- The **Risk Assessment Group (RAG)**, coordinated by Sciensano (**RAG team**) and composed of PH authorities and experts, proposes a risk level and provides recommendations on potential measures → to Risk Management Group (RMG) for validation

SARI Surveillance - outputs : Respi-radar

Week	Consultations GPs for ILI symptoms*	Consultations GPs for ARI*	ILI in nursing homes**	Hospital admissions for SARI*	Concentration SARS-CoV-2 in waste water*****	Evaluation RAG
2023w34	90	462	6	1,9	5	green
2023w35	79	516	7	2,8	10	green
2023w36	104	654	9	0,93	14	green
2023w37	113	641	4	1	7	green
2023w38	101	703	6	1,7	9	green
2023w39	119	1007	6	6,6	14	green
2023w40	128	926	8	2,8	10	green
2023w41	132	920	4	2,1	15	green
2023w42	155	1036	6	1,5	11	green
2023w43	44	845	3	2,9	8	green
2023w44	88	798	7	7	9	green
2023w45	156	1275	6	11	10	green
2023w46	157	990	15	5,2	14	yellow
2023w47	167	989	9	6,8	19	yellow
2023w48	220	1223	11	7,1	20	yellow
2023w49	250	1363	19	9,1	24	yellow
2023w50	275	1271	5	9,4	22	yellow
2023w51	359	1220	8	8,7	27	yellow
2023w52	231	1152	5	13,6	19	yellow
2024w01	234	965	11	12,9	10	yellow
2024w02	283	1140	11	10	11	yellow
2024w03	315	1057	13	10,7	13	yellow
2024w04	537	1363	12	7,9	11	orange
2024w05	532	1406	12	14,6	9	orange
2024w06	523	1176	17	15,2	3	orange
2024w07	267	1070	7	11,3	4	orange
2024w08	230	1093	7	13,3	6	yellow
2024w09	194	925	3	10,1	4	yellow
2024w10	128	890	7	9	1	yellow
2024w11	151	871	7	9,4	2	green
2024w12	62	789	4	9,7	0	green
2024w13	85	909	2	6,7	0	green
2024w14	74	743	5	9,2	0	green
2024w15	56	681	4	8,7	0	green
2024w16	41	679	3	8,1	0	green
2024w17	53	754	1	6,9	0	green

Additional indicators feeding the assessment

- Vaccination
- Mortality
- Qualitative feedback from experts (eg workload)
- Modeling (prediction)
- International and European situation
- Pathogen-specific data, e.g. genomic surveillance SARS-CoV-2

Data available at time of assessment

SARI Surveillance - outputs

- European reporting

- Weekly report through TESSy – The European Surveillance System
- SARISURV: for case-based reporting of SARI cases
- SARISURVDENOM: for reporting of weekly denominators (hospital catchment population and admissions, by age groups)



- European projects

- VEBIS
- PROMISE
- EHR-SARI

Scientific journal article

Vaccine effectiveness against influenza hospitalisation in adults during the 2022/2023 mixed season of influenza A(H1N1)pdm09, A(H3N2) and B circulation, Europe: VEBIS SARI VE hospital network
1 Feb 2024

Scientific journal article

Effectiveness of the adapted bivalent mRNA COVID-19 vaccines against hospitalisation in individuals aged ≥ 60 years during the Omicron XBB lineage-predominant period: VEBIS SARI VE network, Europe, February to August, 2023
1 Jan 2024



PROMISE
Preparing for RSV Immunisation
and Surveillance in Europe



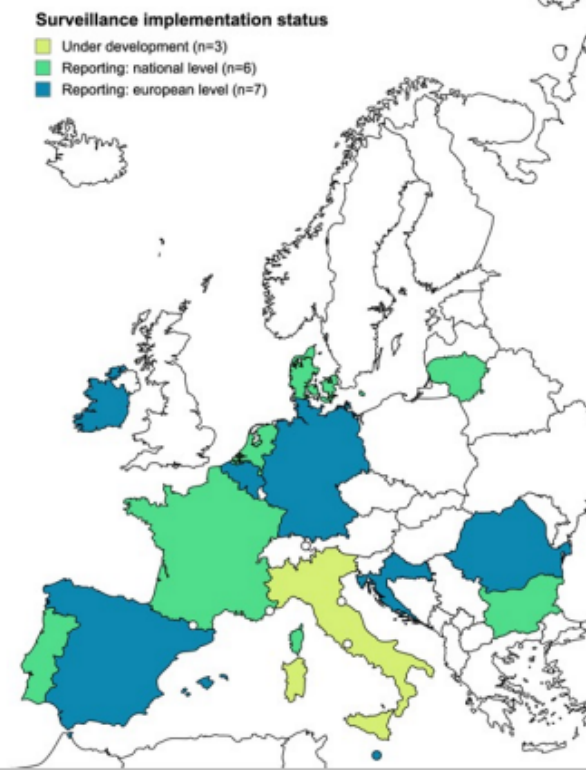
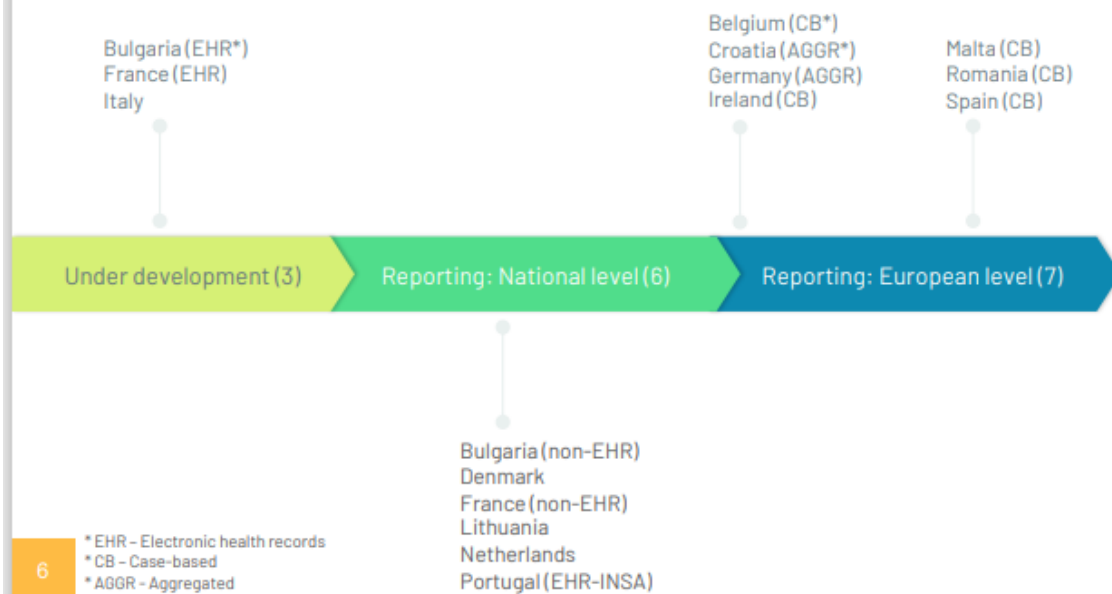
Contribution to European initiatives

SARI Surveillance - outputs

Systems overview

There are different implementation status

- 14 countries with 16 systems
- New countries expected reporting to TESSy in 2024/25: BU, DK, IT, LT, NL



End-of-cycle presentation of EHR-SARI (European project organized by ECDC)

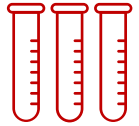
Only 10 countries report SARI data to ECDC

SARI Surveillance - Conclusion



Standardized case identification in a timely manner

- **comparability** and wide selection of cases
- **Weekly** interpretation of the epidemiological situation



Extensive uniform sample testing in a retrospective manner

- **comparability** of test results
- **All-round** interpretation of the epidemiological situation



Systematic validated data collection/processing

- **Increased data quality**
- **Detailed collection** of the patient profiles



Prospective health data integration

- **Pandemic preparedness**
- **linkage** with other datasets



Pooled data assessment

- **Robust** and **consistent** interpretation
- **Comparability over time** (thresholds for epidemic assessment)



Contribution to European Initiatives

Acknowledgements

Respiratory infection surveillance team

Laurane De Mot
Yinthe Dockx
Mathil Vandromme
Sébastien Fierens
Claire Brugerolles
Yves Lafort
Peace Mpakaniye
Adrien Lajot
Sven Hanoteaux

Nathalie Bossuyt

BELSARI research group

<https://www.sciensano.be/en/file/belsari-net-research-group>

Other teams Sciensano

Mortality
RAG
TAS (incl. BePrepared & EpiModelling)
Wastewater
LINK-VACC
VPD
Health services research

Contact

Yinthe Dockx • Yinthe.dockx@sciensano.be